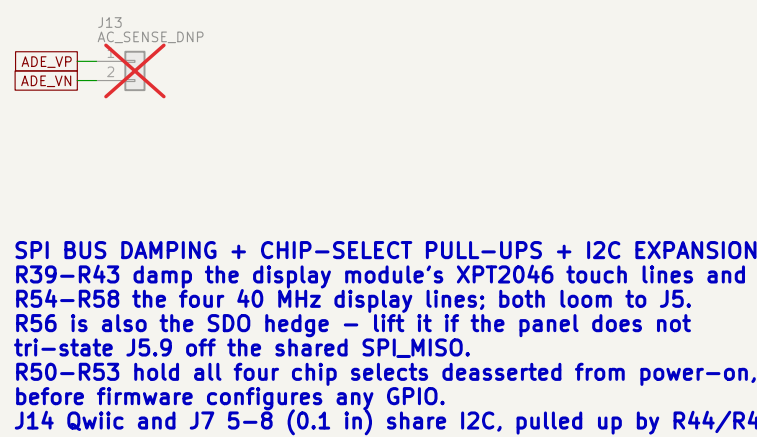
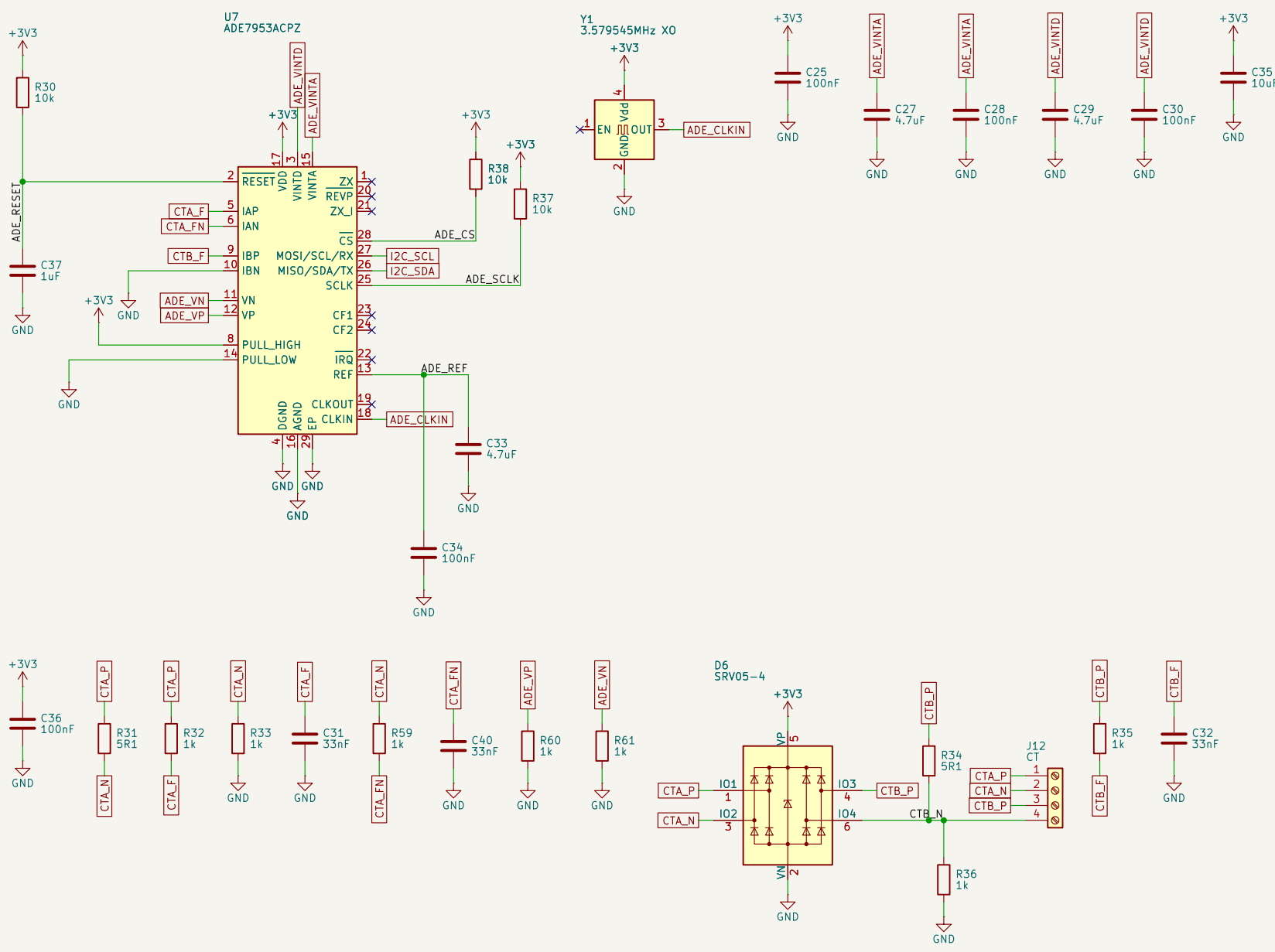
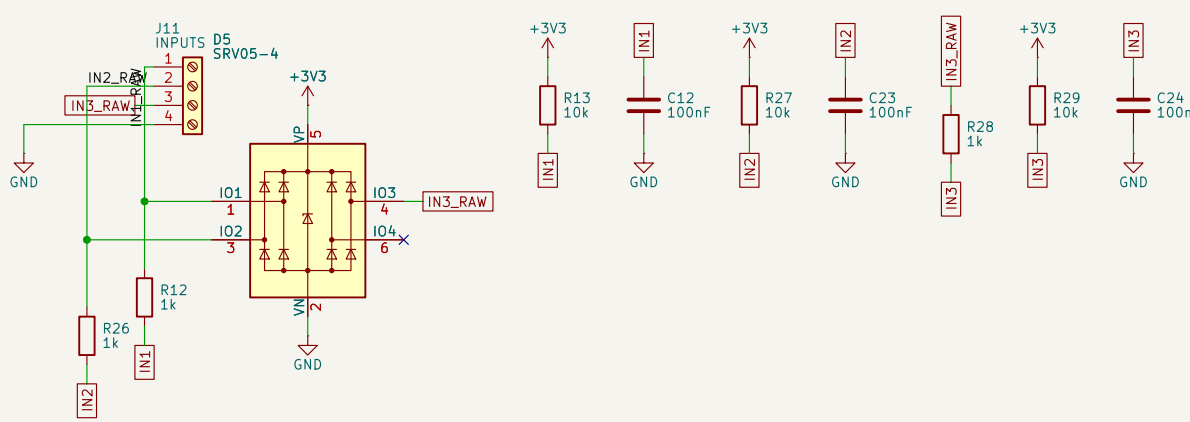
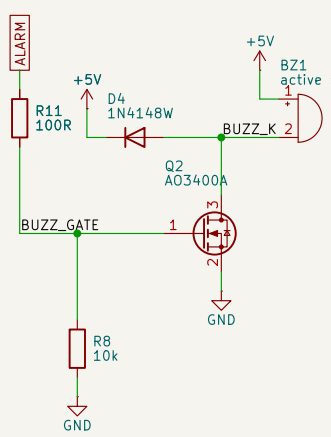
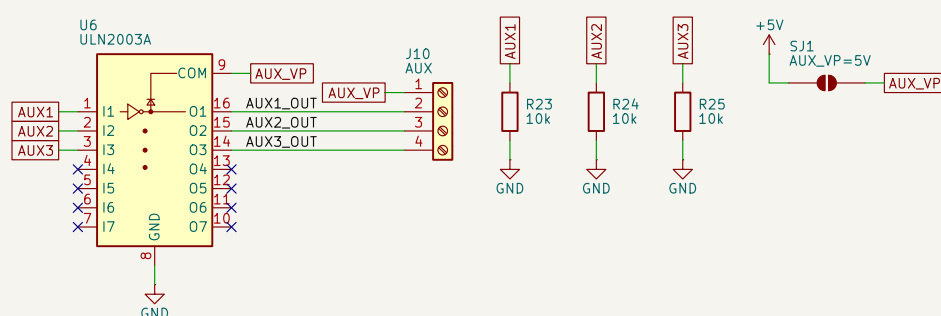
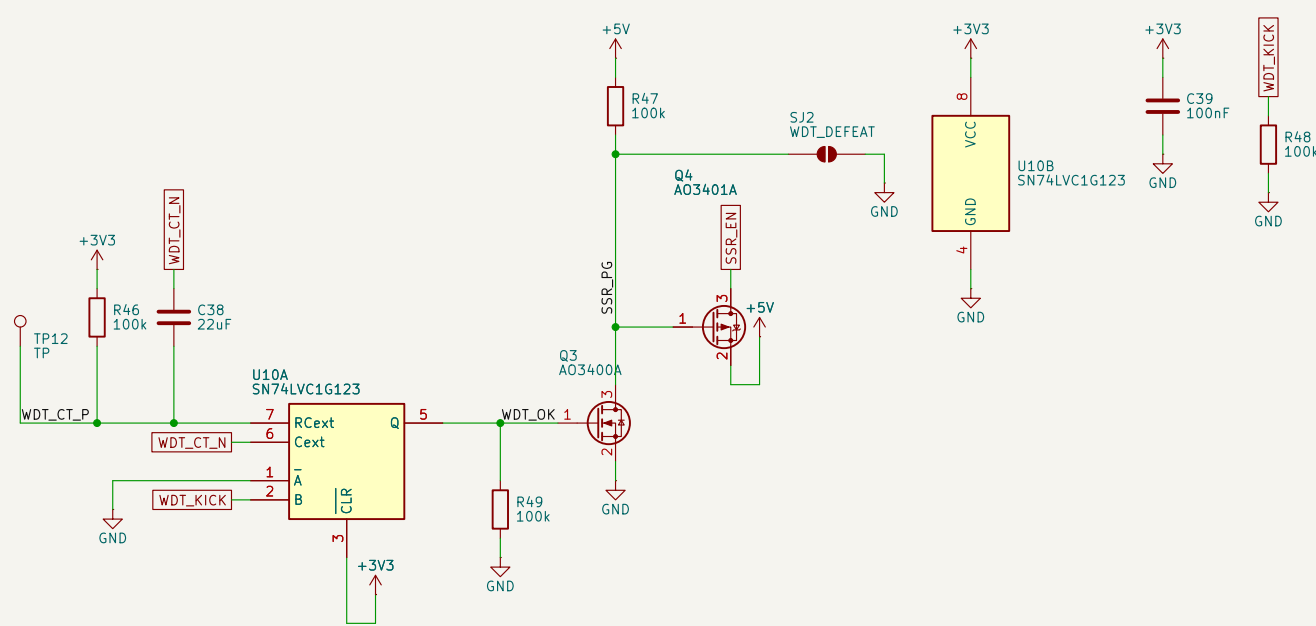
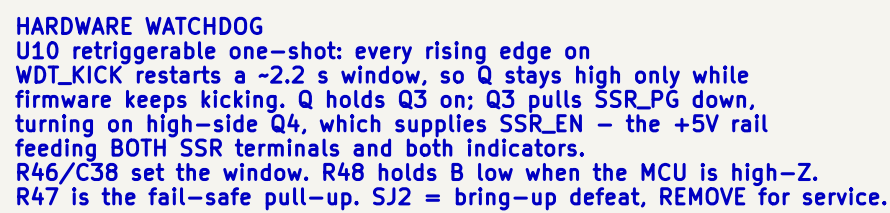
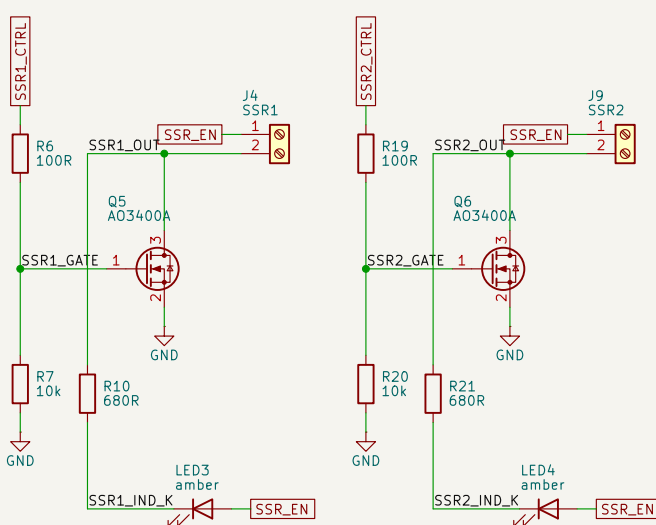
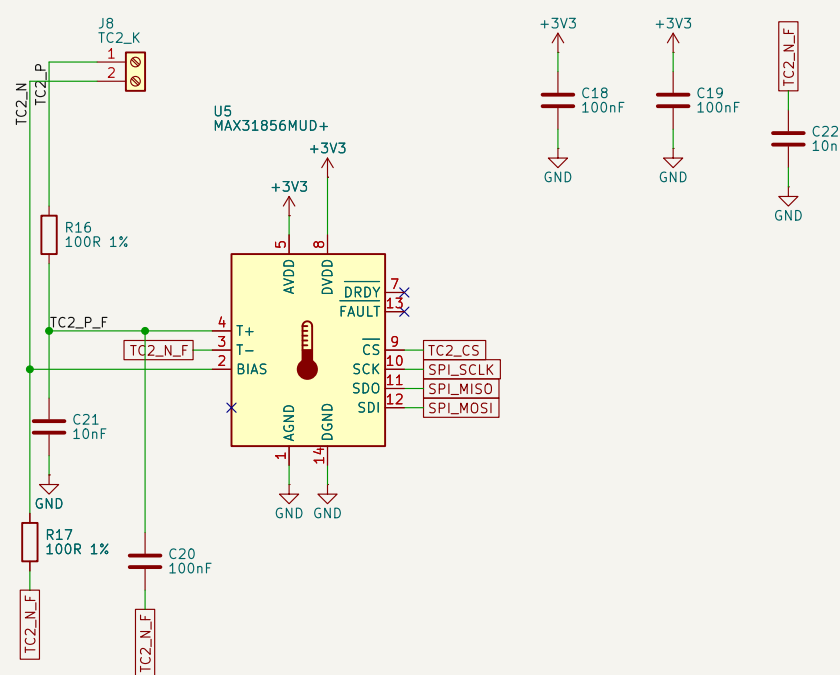
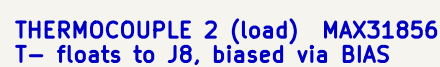
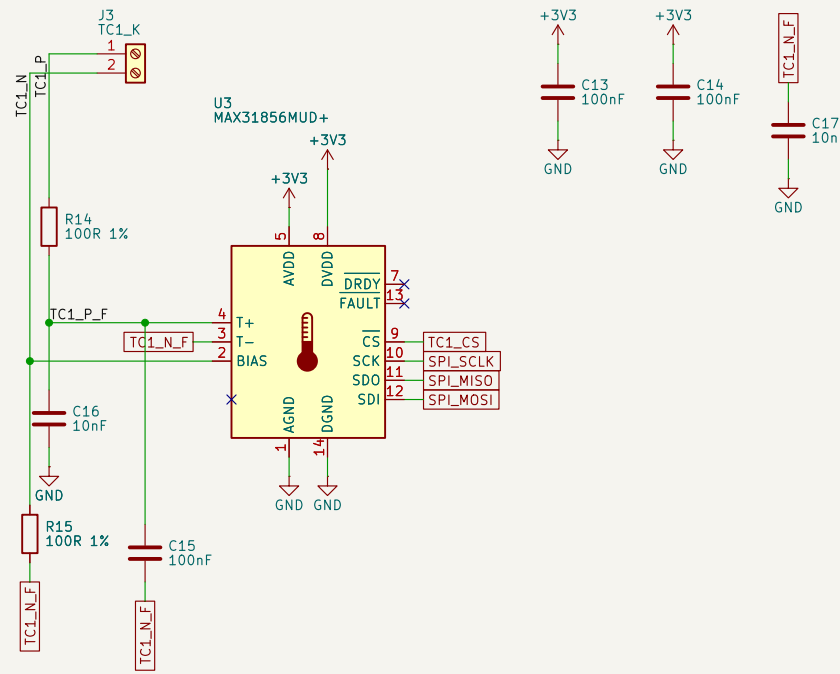
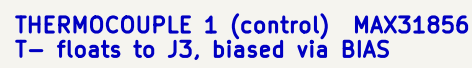
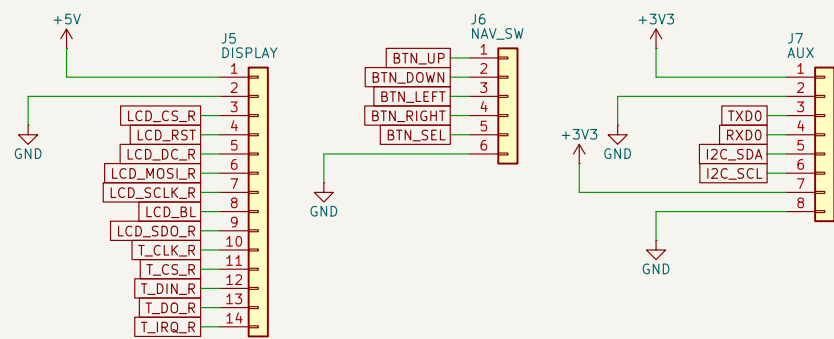
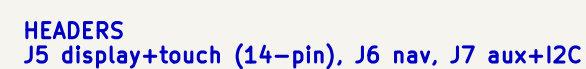
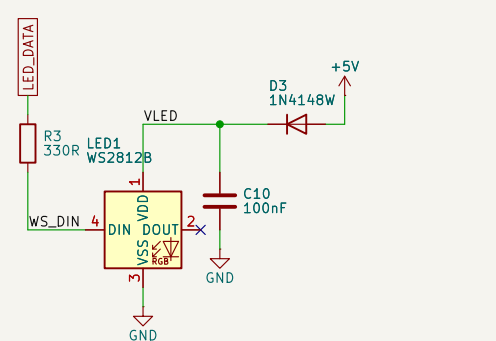
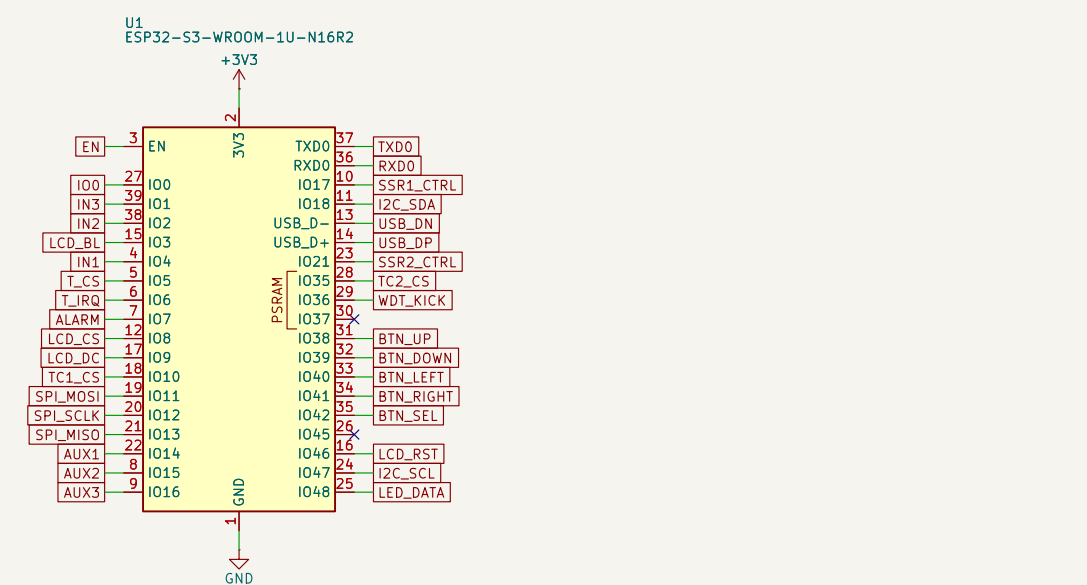
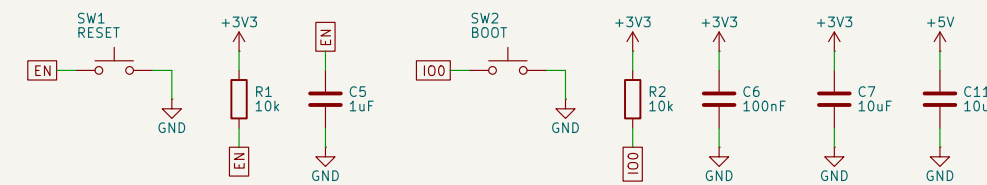
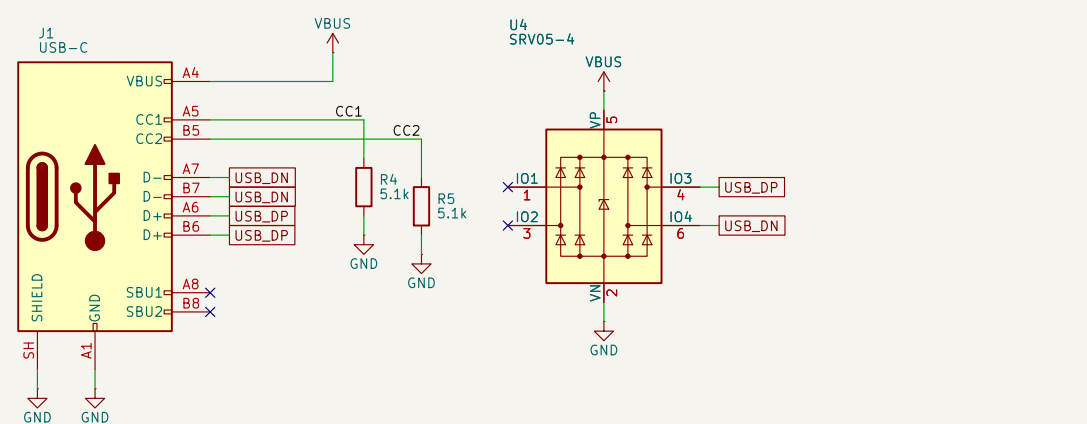
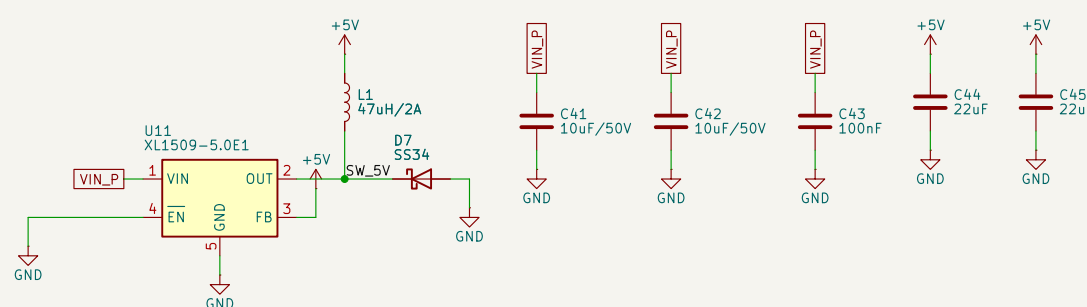
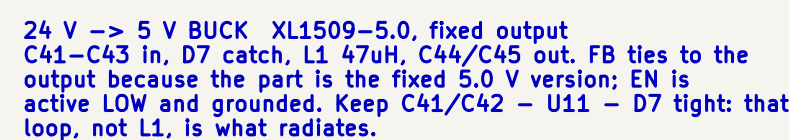
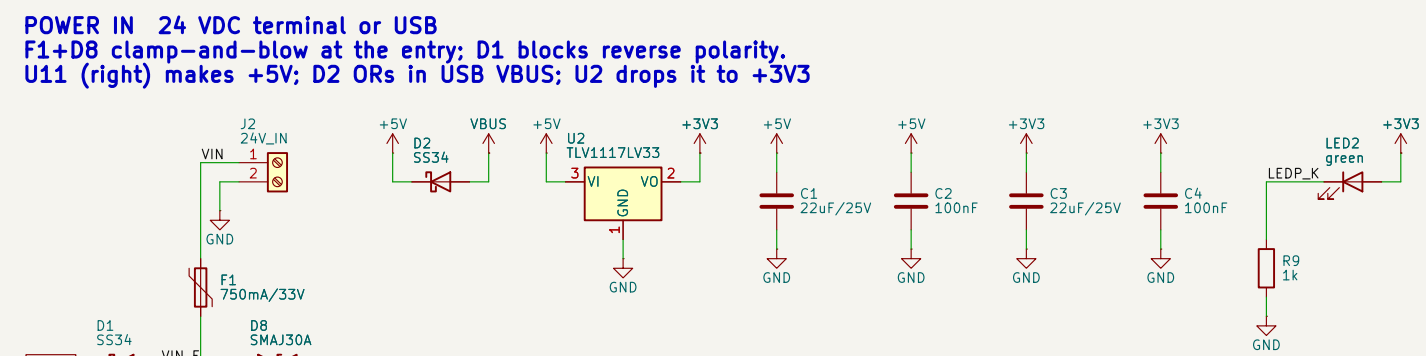


NOTES – Bisque kiln controller, rev E

SSR terminals J19; pin1 = SSR.EN (board +5v, watchdog-gated)
pin2 = the switched low side, Boot+ase; R7/R20 hold each MOSFET gate down while the ESP32 pins are high-impedance.
NOT opto-isolated – rev B tried that and reverted it; an opto only isolates if the SSR loop is powered off-board, and this board powers it.

TC1 terminal J3 / TC2 terminal J8; pin1 = K+, pin2 = K- – both float, biased near AGND only through each MAX31856's internal BIOS network. Ungrounded – junction probes required; two biased – junction probes in one kln would loop through it.

TC1 terminal J3 / TC2 terminal J8: pin1 = K+, pin2 = K- - both float, biased near AGND only through each MAX31856's internal BIAS network. Ungrounded-junction probes required: two grounded-junction probes in one kit would loop through it.

Nav switch J6 is panel-mounted; inputs use ESP32 pull-ups.

J11: IN1/IN2/IN3 (lid/gas-flow/spare) + GND - dry contact each channel to GND.

Display J5 (14-pin, ST7796S + XPT2046 touch module, LCDWIKI MSP430F1): 1=+5V 2=GN 3=CS 4=RS† 5=DC 6=SDI/MOSI 7=SCK 8=BL 9=SDO/MISO 10=TD_CLK 11=TS 12=TD_IN 13=TD_DO 14=TD_IRQ. PIN 1 IS +5V, NOT 3V3 – the module regulates on board; do not wire a 3.3V-only panel to it. Logic is 3.3V; touch shares SPI2 with the LCD and both MAX31856s (R39–R43 damping).

J12: CT current inputs, CTA_P/CTA_N/CTB_P/CTB_N – one CT clamp per SSR zone.

J13: DNP SELV voltage sense header. NOT mains-rated, not fitted - do not wire to AC mains. Y1/C25/C26 load caps are an ASSUMED value (unverified C_L), see design.py comment. ADE7953 I2C address 0x38 collides with PCF8574A.